

COP2073C Module 12 Programming Assignment

For this exercise, you will need to install and load the ggplot2 package.

In this assignment, we will use the ggplot2 package to create a **line chart** that examines the relationship between engine displacement (disp) and miles per gallon (mpg) for cars in the **mtcars** dataset. We will also incorporate the number of cylinders (cyl) to illustrate how different variables interact in influencing fuel efficiency.

A Note on Problem Domains

Are you a car expert? If not, it is essential to research the problem domain before conducting an analysis like this. Understanding terms such as "engine displacement" is crucial for accurate data interpretation and meaningful insights. This knowledge helps you interpret variables correctly, recognize relationships, and avoid misleading conclusions. For example, knowing that larger engine displacement often correlates with changes in fuel efficiency provides context to patterns observed in the data. Domain knowledge also aids in identifying relevant variables, such as the number of cylinders (cyl), and in designing visualizations that effectively highlight significant patterns. It ensures clear communication of findings by framing results in accurate and relatable terms for stakeholders. When data scientists are not familiar with a particular domain, research is required, and consulting domain experts can provide invaluable insights and context, enhancing the relevance and reliability of the analysis.

The mtcars dataset is included in the base R installation, so no additional data loading is necessary. Use the help feature (?mtcars) to find the name and description of the variables.

Instructions:

1. Use ggplot() to set up the base for your line plot, and add layers to make the visualization more informative.
2. Represent:
 - disp on the x-axis
 - mpg on the y-axis
 - cyl as a line grouping factor

Additional Hints:

- Use geom_line() to create the line chart.
- Use color to represent the number of cylinders (cyl). This will help differentiate the data across groups.
- Adjust the alpha parameter to make overlapping lines more distinguishable (e.g., alpha = 0.8).
- Add a descriptive title, subtitle, axis labels, and a caption using labs() to provide context for your plot.
- Use scale_color_discrete() to add a more descriptive legend label for the cylinder variable.
- Position the legend below the plot using theme(legend.position = "bottom") to avoid overlap with the chart.
- Use a minimal theme (theme_minimal()) as a base style to keep the focus on the data, and fine-tune adjustments like text size and alignment using theme().

Save the plot as a PNG file (fuel_efficiency_plot.png)

Write a 150-200 word analysis summarizing the relationships between the variables. Focus on

- How fuel efficiency changes with engine displacement.
- Any trends observed across different cylinder groups.

Non-Functional Requirements:

- Include a 4-line ID header at the beginning of your script.
- Include vertical spacing (a blank line) between logical blocks for readability.
- Comment your code thoughtfully (avoid excessive commenting).
- Ensure each line of code does not exceed 80 columns

Submit your R script, image file, and analysis to the GitHub Classroom repository.